



Principles of Epidemiology

قسم الوبائيات
الفرقة الرابعة



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Introduction to the Book

This textbook, *Epidemiology for Nursing Practice – 2025 Edition*, is designed to provide nursing students with a structured and clinically relevant foundation in the principles of epidemiology. It focuses on the patterns, causes, and prevention of diseases commonly encountered in both hospital and community settings, with emphasis on the Egyptian healthcare context.

This textbook contains 13 chapters covering key topics in epidemiology, including disease classification, transmission routes (foodborne, droplet, bloodborne, contact, zoonotic, waterborne, vector-borne), non-communicable diseases, screening, research methodology, sampling, and epidemiological measurements.

To support students in understanding how epidemiological knowledge is generated, the book introduces basic concepts of research methodology and selecting appropriate study designs. It also explains sampling techniques and their role in ensuring valid and representative data. In addition, the book outlines key principles of measurement in epidemiology, including the use of rates, ratios, and proportions to quantify disease frequency and assess risk.

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Chapter 1: Introduction to epidemiology

Objectives

- Understand the definition and objectives of epidemiology
- Differentiate between descriptive and analytic epidemiology
- Differentiate between Health, disease and illness
- Differentiate between communicable and non-communicable diseases
- Recognize why communicable diseases remain major problem in developing countries
- Recognize theories of disease causation
- Recognize the difference between infection, colonization, contamination, pollution and infestation
- Define dead-end infection and latent infection
- Recognize the difference between eradication, control and elimination

Definition of Epidemiology; it is the study of the distribution and determinants of diseases in the populations, and the application of this study to control health problems.

- **Distribution;** This refers to the pattern of occurrence of disease in the community with reference to time, place and person.
- **Determinants;** This refers to the etiological or risk factors related to the disease.

Objectives of epidemiology

1. To know the distribution of the disease in the community
2. To know the magnitude of the problem
3. To identify the etiological and risk factors in the development of disease

4. To plan for the implementation of prevention and control measures
5. To eliminate or eradicate the disease
6. To evaluate the control measures
7. Ultimate objective is to promote the health and wellbeing of the people.

Difference between Health, Disease, and Illness

- Health is a state of complete physical, mental, and social well-being of an individual and not merely an absence of disease or infirmity.

Physical dimension means all the organs and systems in the body are functioning perfectly at their optimum capacity. A person is said to be mentally healthy, when he/she is having a perfect state of balance with the surrounding world, having harmonious relation with others, the intelligence, memory, learning capacity, judgment is normal, not having any internal conflicts, accepts criticism sportively, has got good self-control. An individual is said to be socially healthy, when he is accepted, respected, and loved by all in the family, by his friends, relatives, neighbors, colleagues and others.

- disease is a physiological or psychological dysfunction of the body.
- Illness is a subjective feeling of not being well

Diseases of man can be broadly classified into communicable and non – communicable diseases.

1. **Communicable** diseases can be **transmitted** from man to man (mainly) or animal to man (sometimes) by different means.

2. Non communicable diseases; are **non-transmissible** disease ex., hypertension, diabetes Miletus and cancer.

Communicable diseases remain major problem in developing countries especially rural and slums areas **due to:**

A- insanitary environment

B- improper health habits and behavior of the public

C- poor quality and underutilized health services.

Theories of disease causation

A variety of theories of disease causation have been proposed. The purpose of these theories is to facilitate the understanding of nature of the disease, which is complex.

1- Old theories:

Till the end of the 18th century, various theories were in vogue, e.g. Supernatural theory of disease (e.g. curse of God; an evil eye).

2- Germ theory:

The discoveries in microbiology at the turn of the 18th century became a turning point in the etiological concept of disease. Louis Pasteur (1860) demonstrated the presence of bacteria in the air. Robert Koch (1877) showed that anthrax was caused by bacteria. These discoveries of Pasteur and Koch confirmed the germ theory of disease. The theory concluded that the disease is caused by single agent (microorganism).

Limitations of Germ Theory

The germ theory is unable to explain:

- a. Why only some people suffer from the disease after exposure to microorganisms and not all, as with exposure to tubercle bacilli.
- b. Why some people do not suffer from the disease even though they harbor the pathogens in the body (as in healthy carriers of typhoid)

3. Epidemiologic triad;

- Agent, host, and environmental factors interrelate to produce disease in humans.
- According to this model (Fig. 1) disease occurs when the equilibrium between agent, host and environment is disturbed. Thus, this model explains that some persons do not suffer from the disease even though they harbor the pathogens because an equilibrium is established between the causative agent and the host.
- This theory mainly linked to **communicable** diseases.

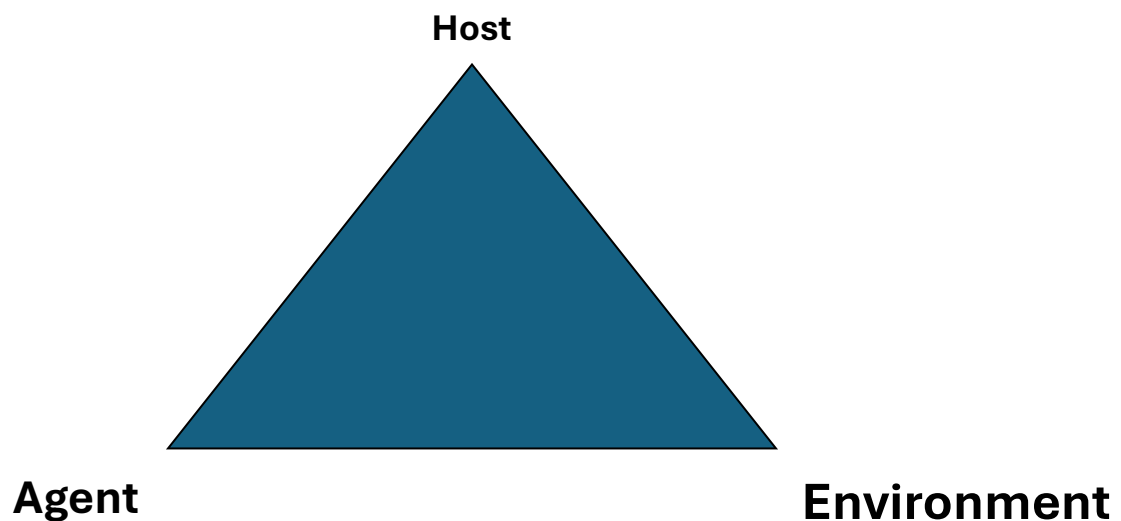


Fig. (1): Epidemiological triad

4. Theory of Multifactorial Causation (Web of Causation)

- A disease is not only caused by an organism but also predisposed by many factors contributing to its occurrence (risk factors).
- most of these risk factors are linked to life-style and human behavior.
- This theory linked mainly to modern diseases of civilization (non communicable diseases)

Important epidemiological terms

1. Infection:

- This refers to the entry and development or multiplication of an infectious agent in the human (or, animal) body, with an implied response (e.g. immunological response) on the part of the human or animal.

2. . Colonization:

- presence of infectious agent in the human body but without any evidence of specific host immune responses to the agent.

3. Contamination:

- Refers to an infectious agent being present in inanimate articles like food, water. This term is also used to denote presence of infectious agent on skin surface, particularly on hands.

4. Pollution:

- Refers to presence of either infectious agent or mechanical agents (as sound), usually in the general environment, air or water (e.g. sound pollution, water pollution, air pollution).

5. Infestation:-

- The presence and development of insect vectors on the body or linen

6. **Dead-End Infection:**

- A state when an infectious disease, which is usually “communicable”, cannot be transmitted any further between human beings or from humans to animals or vice versa, for various agent, host and environmental reasons. Examples are Rabies and Tetanus

7. **Latent Infection :**

- This refers to a state when the infectious agent lies “dormant” within the host body, without any clinical manifestations but does not come out of the human body (thus it is different from subclinical infection).
- After a period of time, under certain circumstances, the agent which had been lying dormant, reactivates (e.g. reactivation Tuberculosis).

8. **Eradication :**

- The term refers to a complete cessation of transmission of the infectious agent.
- The infectious agent as well as the disease has also been completely reduced to zero.
- Small pox is the only example wherein eradication has been achieved.

9. **Control :**

- Reducing the transmission of a disease to a level when it no longer remains a “public health problem”.

10. **Elimination :**

- Elimination implies either a ‘regional eradication’ (say from a country or continent), or else reduction of disease to zero without total removal of infectious agent.

Infectiousness, Infectivity, Pathogenicity and Virulence

- ✓ Infectiousness : this is the relative ease of agent transmission to the host.
- ✓ Infectivity : This is the ability of the agent to enter, survive and multiply in the host.
- ✓ Pathogenicity : It is the ability of the agent to produce manifest disease out of those who have been infected.
- ✓ Virulence : It is the ability of the agent to produce severe disease (fatal).

Exercise

1. Presence of an infectious agent in an inanimate article or on skin surface, particularly hands, is called

- (a) pollution (b) contamination (c) infection (d) infestation

2. The presence and development of insect vectors on the body or linen e.g. louse is known as

- (a) Infection (b) Infestation (c) Infectiousness (d) Infectivity

Chapter 2: Natural history of the disease, and pattern of spread of infectious disease

Objectives

- Recognize the natural history of disease
- Recognize the pattern of spread of infectious diseases

Natural history of disease

- The progress of a disease process in an individual over time, in the absence of intervention fig (2)

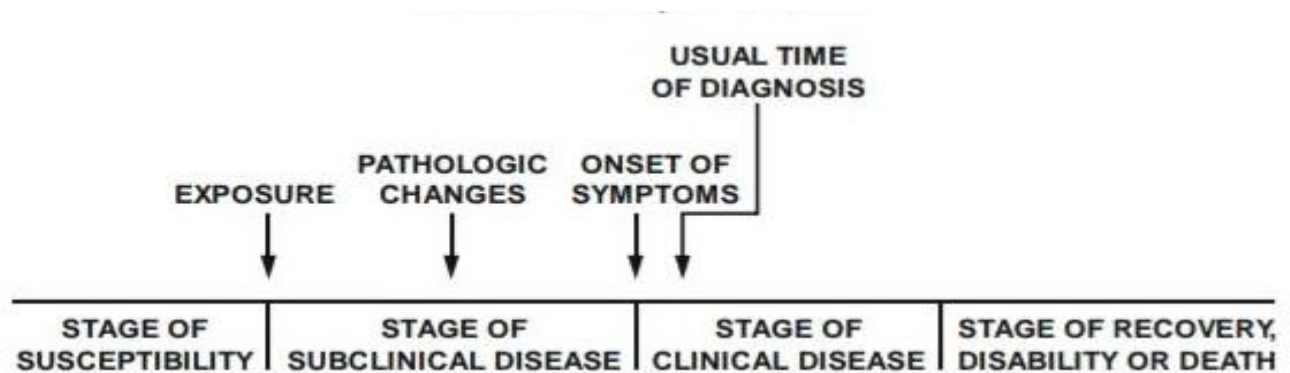


Fig. (2); Natural history of disease

- The natural history begins with the appropriate exposure to or accumulation of factors sufficient to begin the disease process in a susceptible host.
- For infectious disease, the exposure usually is a microorganism.

- For cancers, the critical factors may require both **cancer initiators**, such as asbestos fibers or components in tobacco smoke (for lung cancer) , and **cancer promoters**, such as estrogens (for endometrial cancer).

▪ **Subclinical Infection (Inapparent infection) :**

It is a state when there is a host immune response following entry of the infectious agent; the agent may also multiply in the host body, but there are no clinical manifestations of the disease.

- Although disease is inapparent during the incubation period, some pathologic changes may be detectable with laboratory, radiographic, or other screening methods.
- Most screening programs attempt to identify the disease process during this phase of its natural history, since early intervention may be more effective than treatment at a later stage of disease progression.

▪ **Incubation or latency period**

Usually, a period of subclinical or inapparent pathologic changes follows exposure, ending with the onset of symptoms.

- ✓ **For infectious diseases**, this period is usually called the **incubation period**
- ✓ **for chronic diseases**, this period is usually called **the latency period**



- **Stage of clinical disease**

- The onset of symptoms marks the transition from subclinical to clinical disease. Most diagnosis is made during the stage of clinical disease.
- In some people, however, the disease process may never progress to clinically apparent illness. In others, the disease process may result in a wide spectrum of clinical illness, ranging from mild to severe or fatal.

- **Stage of recovery, disability or death**

- The infectious diseases have various outcomes according to the infectivity, pathogenicity and virulence of the causative agent.
 - a) Infectivity refers to the proportion of exposed persons who become infected.
 - b) Pathogenicity refers to the proportion of infected persons who develop clinical disease.
 - c) Virulence refers to the proportion of persons with clinical disease who become severely ill or die.
- For example, hepatitis A virus in children has low pathogenicity and low virulence, since many infected children remain asymptomatic, and few develop severe illness.
- Rabies virus is both highly pathogenic and virulent, since virtually 100% of all infected persons (who do not receive treatment) progress to clinical disease and death.

Pattern of Spread of infectious diseases

The amount of a particular disease that is usually present in a community is the baseline level of the disease. This level is not necessarily the preferred level, which should in fact be zero; rather it is the observed level. Theoretically, if no intervention occurred and if the level is low enough not to deplete the pool of susceptible persons, the disease occurrence should continue at the baseline level indefinitely. Thus, the baseline level is often considered the expected level of the disease.

3- Endemic spread:

- a disease constantly present in the community due to maintained infection by existing ecological factors (agent, host, and environment).

Different diseases, in different communities, show different patterns of expected occurrence:

- a persistent level of occurrence with a low to moderate disease level is referred to as an endemic level. Ex., Bilharziasis is endemic in Egypt.
- a persistently high level of occurrence is called a hyperendemic level.

- an irregular pattern of occurrence, with occasional cases occurring at irregular intervals is called sporadic.

4- **Epidemic spread:**

- Increase number of cases significantly more than the usual pattern of spread of the disease. Epidemic is characterized that all the cases are interrelated :
- Time: all cases appear within a certain period of time.
- Place: all cases are reported within the same area or community.

Epidemic is either :

5- endemic infectious disease in a community and show epidemic wave at any time under adverse circumstances. E.g., Typhoid fever

B- some infectious diseases not known before or have been eliminated may appear or reappear and show epidemic wave .e.g. cholera

An epidemic may result from the following:

- a. a recent increase in amount or virulence of the agent
- b. the recent introduction of the agent into a setting where it has not been before
- c. an enhanced mode of transmission so that more susceptibles are exposed
- d. some change in the susceptibility of the host response to the agent
- e. factors that increase host exposure or involve introduction through new portals of entry

Outbreak spread: is a localized epidemic that involves a confined group or closed community as camp, school, nursery.

6- **Pandemic spread:**

- an epidemic of a particular infectious disease that spreads in between countries .e.g., COVID-19

7- **Enzootic spread:**

- endemic spread of infectious diseases in animals with the potential risk of transmission to man . e.g., tuberculosis

8- **Epizootic:**

- epidemic spread of infectious diseases in animals with the potential risk of transmission to man .e.g., foot and mouth disease.

Exercise; The case-report pattern of disease X for three communities. The communities have the same size population. Identify which term A-D below best describes the occurrence of disease X.

A. Endemic B. Epidemic C. Hyperendemic D. Pandemic

1. Community A: usually 10 cases/week; last week, 28 cases (....)

2. Community B: 50-70 cases/week; last week, 55 cases (....)

3. Community C: usually 25 cases/week; last week, 28 cases (....)

Chapter 3: The chain of infection

Objectives

- Understand the chain of infection.
- Recognize the epidemiological importance of human carriers

As described before, the traditional epidemiologic triad model holds that infectious diseases result from the interaction of agent, host, and environment. More specifically, transmission occurs when the agent leaves its reservoir or host through a portal of exit, is conveyed by some mode of transmission, and enters through an appropriate portal of entry to infect a susceptible host. This sequence is sometimes called the chain of infection.

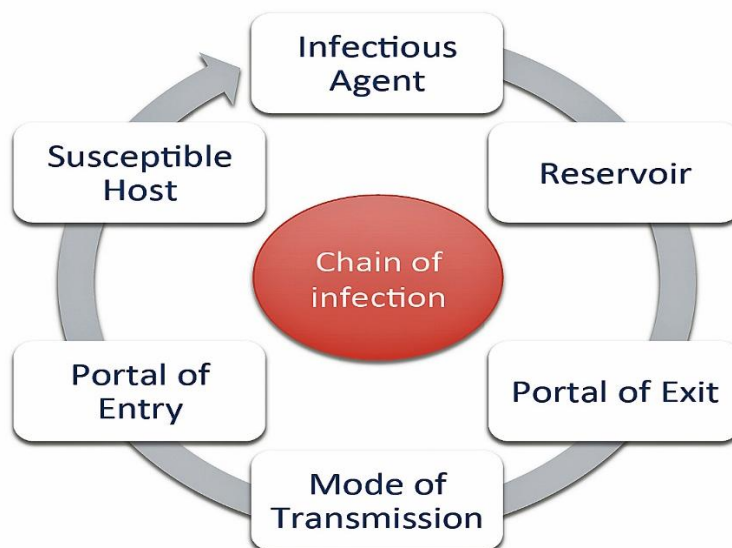


Fig. (3) : The chain of infection

Reservoir

The reservoir of an infectious agent is the habitat in which the agent normally lives, grows, and multiplies. Reservoirs include humans, animals, and the environment. The reservoir may or may not be the source from which an agent is transferred to a host. For example, the reservoir of *Clostridium botulinum* is soil, but the source of most botulism infections is improperly canned food containing *C. botulinum* spores.

Human reservoirs. Many common infectious diseases have human reservoirs. Diseases that are transmitted from person to person without intermediaries include the sexually transmitted diseases, measles, mumps, streptococcal infection, and many respiratory pathogens. Because humans were the only reservoir for the smallpox virus, naturally occurring smallpox was eradicated after the last human case was identified and isolated

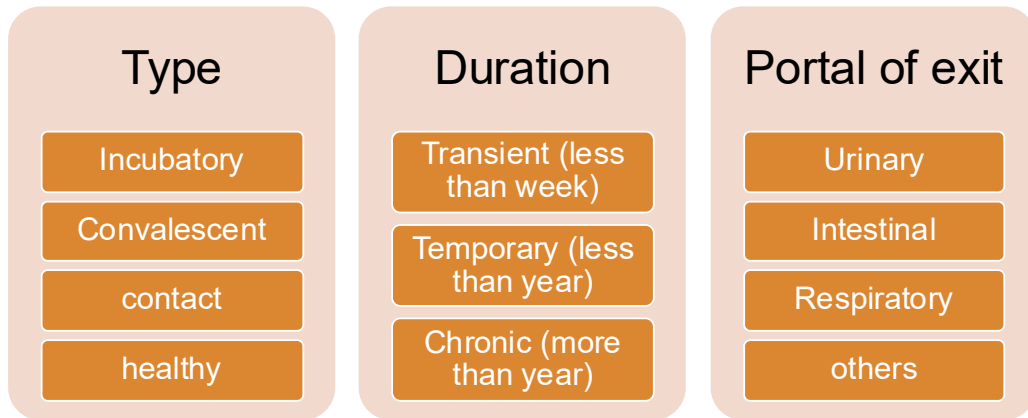
Human reservoirs include:

- Cases: patients show manifestation of the disease.
- Carriers: a carrier is apparently healthy individual who is **infected and harbors the pathogenic organism as a foci of infection** in different parts of his body. *Fig (4)*.

Carrier has an epidemiological importance

- 9- He doesn't show manifestations of disease, but he can spread infection.
- 10- pass free in the community
- 11- diagnosis needs laboratory investigations
- 12- the danger of carrier in some occupations dealing with community members: food handlers, nurses, teachers.

Carriers



6

Fig. (4): Human carriers

- Incubatory carrier: some cases is infectious in its incubation period.
- Convalescent carrier : a certain percent of convalescents (recovered) cases continues to harbor the causative organism in particular foci for varied period of time (weeks, months, years or lifelong)
- Contact carrier: contacts of cases or carriers of some infectious diseases may get infected but not diseased. Infection is eliminated in around two weeks.
- Healthy carrier: inhabitants of endemic areas of infectious diseases may get infected from polluted environment; water supply and food. Infection is eliminated in two weeks.

Animal reservoirs. Humans are also subject to diseases that have animal reservoirs. Many of these diseases are transmitted from animal to animal, with

humans as incidental hosts. The term zoonosis refers to an infectious disease that is transmissible under natural conditions from vertebrate animals to humans. Long recognized zoonotic diseases include brucellosis (cows and pigs), anthrax (sheep), plague (rodents), trichinellosis/trichinosis (swine), tularemia (rabbits), and rabies (bats, raccoons, dogs, and other mammals).

Environmental reservoirs. Plants, soil, and water in the environment are also reservoirs for some infectious agents. Many fungal agents, such as those that cause histoplasmosis, live and multiply in the soil. Outbreaks of Legionnaires disease are often traced to water supplies in cooling towers and evaporative condensers, reservoirs for the causative organism *Legionella pneumophila*.

Portal of exit

Portal of exit is the path by which a pathogen leaves its host. The portal of exit usually corresponds to the site where the pathogen is localized. For example, influenza viruses and *Mycobacterium tuberculosis* exit the respiratory tract, schistosomes through urine, cholera vibrios in feces, *Sarcoptes scabiei* in scabies skin lesions, and enterovirus 70, a cause of hemorrhagic conjunctivitis, in conjunctival secretions. Some bloodborne agents can exit by crossing the placenta from mother to fetus (rubella, syphilis, toxoplasmosis), while others exit through cuts or needles in the skin (hepatitis B) or blood-sucking arthropods (malaria). Fig (5).



Fig. (5): Exit of infection

Modes of transmission

An infectious agent may be transmitted from its natural reservoir to a susceptible host in different ways. There are different classifications for modes of transmission. Here is one classification:

1. Direct
 - a. Direct contact (skin-to-skin contact, kissing, and sexual intercourse)
 - b. Droplet spread (fig. 6)
2. Indirect
 - a) Airborne (fig. 7)
 - b) Vehicle borne (food, water, biologic products (blood), and fomites (inanimate objects such as handkerchiefs, bedding, or surgical scalpels). A vehicle may passively carry a pathogen as food or water may carry hepatitis A virus. Alternatively, the vehicle may provide an environment in

which the agent grows, multiplies, or produces toxin as improperly canned foods provide an environment that supports production of botulinum toxin by *Clostridium botulinum*.

- c) Vector borne. Vectors such as mosquitoes, fleas, and ticks may carry an infectious agent through purely mechanical means or may support growth or changes in the agent. Examples of mechanical transmission are flies carrying *Shigella* on their appendages and fleas carrying *Yersinia pestis*, the causative agent of plague, in their gut. In contrast, in biologic transmission, the causative agent of malaria or guinea worm disease undergoes maturation in an intermediate host before it can be transmitted to humans

Another classification of modes of transmission of infection

1. Droplet infection

- a) Droplet transmission (fig. 6)
- b) Airborne transmission (fig. 7)
- c) Dust: big droplets fall on the floor, get dried including the organisms and rise again in air with dust
- d) Milk : contaminated by 2 respiratory pathogens and invade upper respiratory mucosa : haemolytic *Streptococci* and diphtheria bacilli.

Droplet transmission

Characteristics of droplet particles

- **Size:** large particles (> 5 microns)
- Note: One droplet contains many viruses
- **Behavior:** succumb to gravity, meaning they fall down after travelling in the air
 - **Travel Distance:** short distances, usually land within 1 to 2 meters.



T-zone

Transmission process

- Patient can transmit infection by coughing, sneezing
- We can get infected when droplets contact directly or indirectly mucous membranes of our eyes, nose, or mouth (T-zone).

Fig. 6: Droplet transmission.

Airborne transmission

Characteristics of aerosols particles

- **Size:** tiny particles (≤ 5 microns)
 - **Behavior:** remain active in the air over time and distance
 - **Travel Distance:** can spread over long distances, depending of initial velocity, airflow, and humidity.
- Note: infected person does not need to be in the same room than others to infect them.



Respirator

Transmission process

- Patient can transmit infection by coughing, sneezing, talking, exhaling or when an Aerosol Generated Procedure (AGP) is performed
- We can get infected when inhaling these tiny infected particles. Hence the importance of using a fit-tested respirator.

Fig. 7: Airborne transmission.

2- Ingestion (feco-oral transmission):

by contaminated hands, food or water (fig. 8).

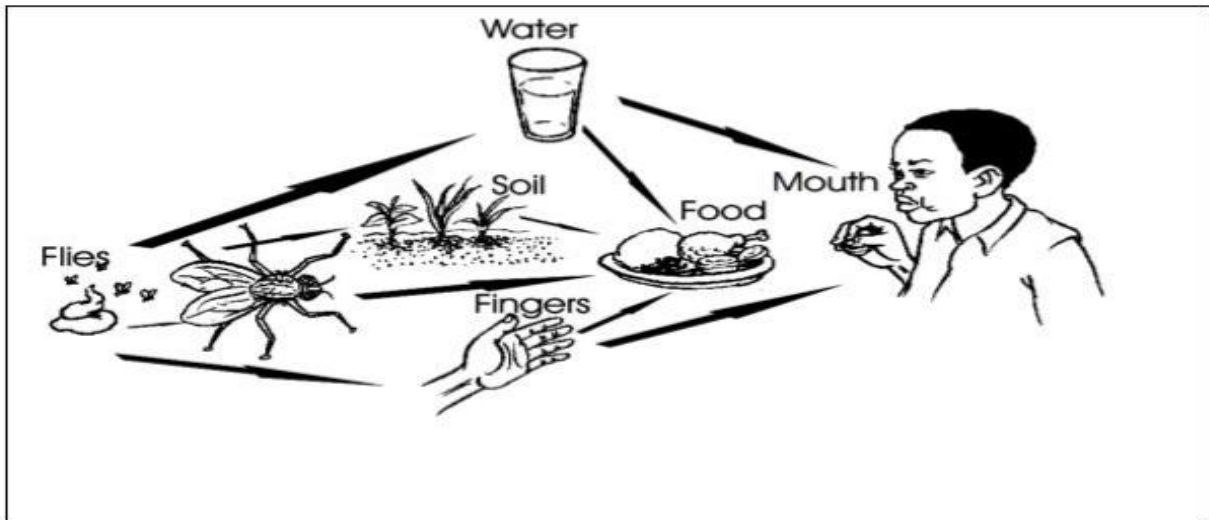


Fig. (8): Feco-oral transmission

3- arthropod- borne infections: mosquitos, fleas , cockroaches.

4- contact infection: invasion of skin and mucus membrane by pathogenic organisms.

5- blood: HBV , HCV, HIV, Malaria

6- vertical : from mother to infant (during pregnancy labor or breast feeding)

Portal of entry

The portal of entry refers to the manner in which a pathogen enters a susceptible host. The portal of entry must provide access to tissues in which the pathogen can multiply or a toxin can act. Often, infectious agents use the same portal to enter a new host that they used to exit the source host. For example, influenza virus exits the respiratory tract of the source host and enters the respiratory tract of the new host. In contrast, many pathogens that cause gastroenteritis follow a so-called “fecal-oral” route because they exit the source host in feces, are carried on inadequately washed hands to a vehicle such as food, water, or utensil, and enter a new host through the mouth.

Exposed host

The final link in the chain of infection is a susceptible host. Susceptibility of a host depends on genetic or constitutional factors, specific immunity, and nonspecific factors that affect an individual's ability to resist infection or to limit pathogenicity. An individual's genetic makeup may either increase or decrease susceptibility. For example, persons with sickle cell trait seem to be at least partially protected from a particular type of malaria. Specific immunity refers to protective antibodies that are directed against a specific agent. Such antibodies may develop in response to infection, vaccine, or toxoid (toxin that has been deactivated but retains its capacity to stimulate production of toxin antibodies) or may be acquired by transplacental transfer from mother to fetus or by injection of antitoxin or immune globulin. Nonspecific factors that defend against infection include the skin, mucous membranes, gastric acidity, cilia in the respiratory tract, the cough reflex, and

nonspecific immune response. Factors that may increase susceptibility to infection by disrupting host defenses include malnutrition, alcoholism, and disease or therapy that impairs the nonspecific immune response.

Knowledge of the chain of infection provides a basis for determining appropriate control measures.

Control measures are directed at:

- Controlling or eliminating agent at source of transmission (interrupt vector borne transmission, measures may be directed toward controlling the vector population, such as spraying to reduce the mosquito population)
- Protecting portals of entry (bed nets are used to protect sleeping persons from being bitten by mosquitoes that may transmit malaria.)
- Increasing host's defenses (Vaccinations promote development of specific antibodies that protect against infection. On the other hand, prophylactic use of antimalarial drugs, recommended for visitors to malaria-endemic areas, does not prevent exposure through mosquito bites, but does prevent infection from taking root)

Chapter 4: levels of prevention

There are three levels of prevention—primary, secondary and tertiary

Primary prevention

This is the measure undertaken in the period of **pre pathogenesis (i.e. before the development of the disease)** which removes the possibility of occurrence of the disease. Especially, it is all the more important in those diseases, for which no treatment is available, e.g. AIDS, cancer, rabies, caries tooth, etc.

Primary prevention can be adopted by two modes of intervention—namely Health promotion and Specific protection.

Health Promotion

This consists of ‘general measures’, which will strengthen the individual/host and prevents the occurrence of the disease by interrupting the interaction among the three factors of epidemiological triad. Examples;

- Health education
- Adequate nutrition
- Improvement in the environmental sanitation (such as control of insects, provision of protected water supply, sanitary disposal of sewage, etc

Specific Protection

This consists of ‘specific measures’, which prevent specific diseases. Examples;

- Immunization against vaccine preventable diseases

- Silver nitrate or penicillin eye drops against ophthalmia neonatorum
- Condom against AIDS
- Helmet against head-injury

Secondary prevention

- This is the measure undertaken in the early stage, **after the onset of the disease (in the early pathogenesis phase).**
- The intervention is by **‘Early diagnosis and treatment ex.,**
 1. Periodic recording of a weight of the child to detect malnutrition
 2. Screening of blood donors to detect HIV/AIDS, hepatitis B, syphilis, malaria
 3. Recording of blood pressure to detect hypertension.

Different modalities of treatment are chemotherapy, radiotherapy, immunotherapy, hormonal therapy, psychotherapy, physiotherapy, oral rehydration therapy and surgery. Conditions where the treatment is for a long period as in tuberculosis or leprosy, the physician should ensure ‘Case holding’, i.e. to see that the patient takes the treatment correctly and completely.

Tertiary prevention

- This is the measure undertaken, **when the disease process is sufficiently advanced, (i.e. in the late pathogenesis phase).**
- It includes two modes of intervention; Disability limitation__ and Rehabilitation.

1. Disability Limitation:- This means limiting the development of further disability in the individual by giving intensive or aggressive treatment

2. Rehabilitation:-

- Combined and use of physical, social, vocational and psychological measures for retraining the individual to the highest possible level of functional ability', so that the individual becomes useful to himself, to the family and to the community at large.
- Physical rehabilitation→ ex., artificial limb.
- Vocational rehabilitation → ex., employment
- Social rehabilitation → Restoration of relationship in the society
- Psychological rehabilitation → Restoration of personal confidence

The sequence of events in a disease process are:

➤ ***Disease → Impairment → Disability → Handicap***

Disease or event (accident)→ Impairment (loss of organ structure or function e.g., loss of foot) → Disability (Inability to perform normal daily activities e.g., inability to walk) → Handicap (loss of social role e.g., Loss of job or unemployment)

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Chapter 5: screening

Objectives

- *Define screening.*
- *Identify aims and objectives of screening.*
- *Know the differences between screening and diagnostic tests.*
- *Identify types of screening.*
- *Recognize the criteria required to launch a screening program.*
- *Identify measures for evaluation of screening test.*
- *Define and calculate sensitivity, specificity, and yield of the test*

Definition of screening

The search for unrecognized disease or defect by means of rapidly applied tests, examinations or other procedures to sort out apparently well persons who probably have a disease from those who probably do not. Screening is used for case detection, control of disease and research purposes



Fig. (9): Screening

- In passive case finding where patients come to medical personnel with symptoms of disease, they are examined carefully and investigated to be diagnosed.
- On the other hand, in screening, the medical team examines persons without symptoms to detect risk factors or early subclinical disease.

Aims and objectives

- The basic purpose of screening is to sort out from a large group of apparently healthy persons those likely to have the disease or at increased risk of the disease under study.
- Screening is carried out in the hope that earlier diagnosis and subsequent treatment favorably alters the natural history of the disease in a significant proportion of those who are identified as "positive".

▪ Differences between screening and diagnostic tests

▪ Screening test	▪ Diagnostic test
▪ Done on apparently healthy	▪ Done on those with disease indication
▪ Used on groups	▪ Used on individual basis
▪ Less accurate	▪ More accurate
▪ Less expensive	▪ More expensive
▪ Not a basis for treatment	▪ Used as a basis for treatment

Types of screening

1. Mass screening:

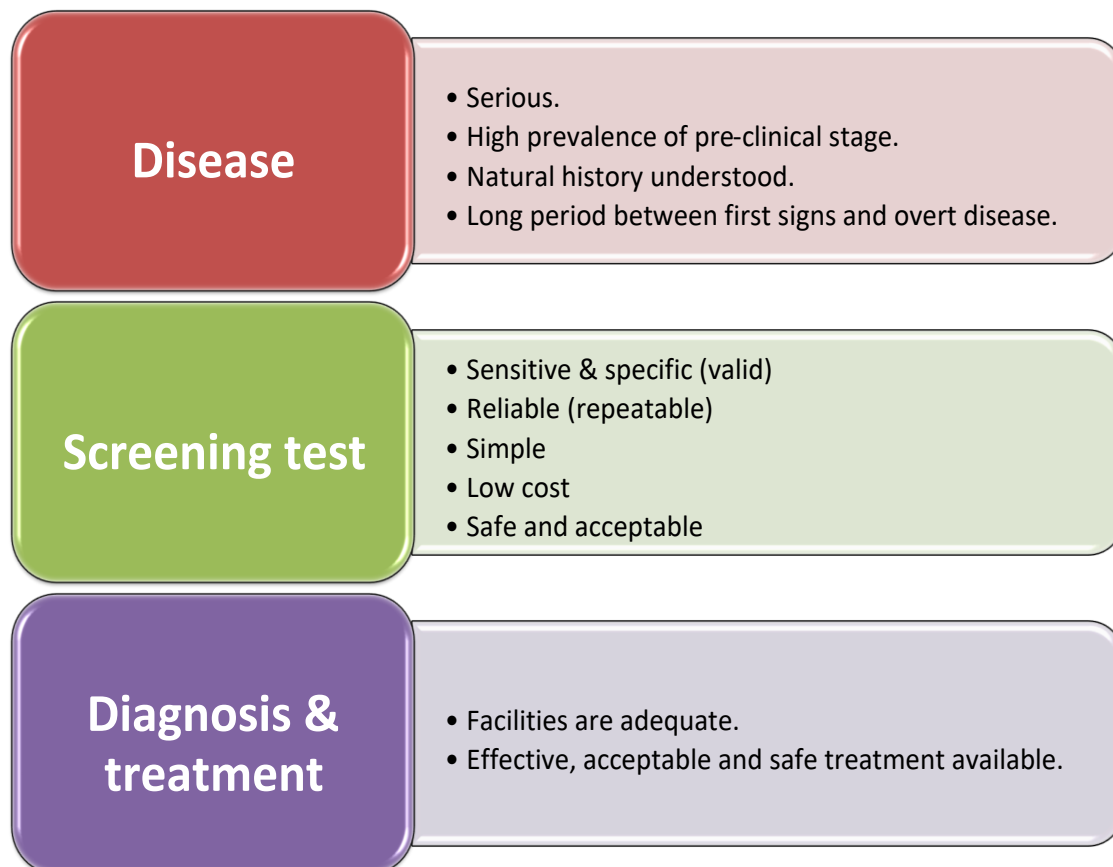
Mass screening simply means the screening of a whole population.

2. High risk or selective screening:

3. Multi-phasic screening:

The application of two or more screening tests in combination to a large number of people at one time.

Criteria for instituting a screening program



Evaluation of screening test

Screening can be evaluated by:

- I. Repeatability (or Reliability) of a screening test.
- II- Validity (or accuracy) of screening test.
- III- Yield of the test.

I. Repeatability (or Reliability) of a screening test.

- A repeatable test is the test that gives the same results if it is repeated many times on the same individual under the same conditions, so repeatability is the percent agreement between multiple results of the test.

Variations in results of the test mean poor repeatability. It may be due to:

- 1- Variation in the characteristic being measured.
- 2- Variation due to observer (tester).
- 3- Variations in instruments or methods.

Measures to enhance repeatability:

- 1- Clear operational definition of variables or cases.
- 2- Good training of data collectors.
- 3- Standardization of questions and tests.
- 4- Data collectors or observers should not be changed in midstream.
- 5- Use of good and well-calibrated instruments.

II- Validity (or accuracy) of screening test

- Validity expresses the ability of a test to separate or distinguish those who have the disease from those who do not.
- It is the degree to which the test measures what is intended to measure.
- It is assessed by comparing the results of the used test with an already accepted test (Valid test).

Sensitivity:

- **The probability of the screening test to be positive when the disease is truly present**

$$\text{Sensitivity} = \frac{\text{True positive (a)}}{\text{True positive (a) + False negative (c)}} \times 100$$

$$\text{Sensitivity} = \frac{\text{Diseased and positive by the screening test}}{\text{Total diseased (validating test)}} \times 100$$

	Disease or Condition	No Disease or Condition
Test Positive	A True Positive	B False Positive
Test Negative	C False Negative	D True Negative

Specificity:

- The probability of the screening test to be negative when the disease is truly absent.

$$\text{Specificity} = \frac{\text{True negative (d)}}{\text{True negative (d) + False positive (c)}} \times 100$$

$$\text{Sensitivity} = \frac{\text{Not diseased and negative by the screening test}}{\text{Total not diseased (validating test)}} \times 100$$

The best screening test is one with high sensitivity and specificity.

Exercise

Valid test screening	<u>ill</u>	Not-ill	
Positive	40	20	60
Negative	60	80	140
	100	100	200

❑ Sensitivity = $40/100 = 40\%$

❑ Specificity = $80/100 = 80\%$

- In a cancer cervix screening program, the screening test was positive in 1300 women and negative in 18700 women. Screening test was positive only in 300 of the 400 confirmed cases.

	ill	Not ill	total
Positive	300	1000	1300
negative	100	18600	18700
total	400	19600	20000

❑ Sensitivity = $300/400 = 75\%$

❑ Specificity = $18600/19600 = 95\%$

III- Yield of the test

- This is the amount of previously unrecognized disease, which is diagnosed and brought to treatment as a result of screening. It depends on sensitivity, specificity and prevalence of the disease.
- The value of screening program is ultimately determined by its effect on morbidity, mortality and disability.

Exercise

1. which of the following is required for enhancing repeatability of the test
 - a) clear operational definition of variables or cases
 - b) variation of the tests used
 - c) changing of data collectors in the mid stream
 - d) using poor calibrated instruments
2. The probability of the screening test to be positive when the diseases is truly present is
 - a) sensitivity
 - b) specificity
 - c) reliability
 - d) yield

Chapter 6: Food borne Diseases

Intended Learning Objectives

By the end of this chapter, students should be able to:

1. Define foodborne diseases and distinguish between infection and intoxication.
 2. Classify foodborne diseases based on causative agents and mechanisms.
 3. Describe the clinical features and transmission routes of key pathogens including Cholera, Poliomyelitis, Hepatitis A, Typhoid and Brucellosis.
 4. Recognize the role of food handlers and environmental factors in disease spread.
 5. Recognize preventive measures including vaccination to reduce the risk of foodborne outbreaks.
-

1. Overview of Foodborne Diseases

Foodborne diseases result from ingestion of food contaminated with infectious agents or their toxins. These may be classified as:

- **Food Infection:** Caused by ingestion of live pathogens that multiply in the host.
- **Food Intoxication:** Caused by ingestion of preformed toxins in food.

They represent a major public health concern, especially in developing countries where food safety systems may be under-resourced.

2. Classification of Foodborne Diseases

Category	Examples
Bacterial Infections	<i>Salmonella, Shigella, E. coli, Listeria, Brucella, Vibrio cholerae</i>
Bacterial Intoxications	<i>Staphylococcus aureus, Clostridium botulinum, Bacillus cereus</i>
Viral Infections	Hepatitis A virus, Norovirus, Poliovirus
Parasitic Infections	<i>Giardia lamblia, Entamoeba histolytica, Taenia saginata</i>
Chemical Poisoning	Pesticides, heavy metals, food additives
Fungal Toxins	Aflatoxins from <i>Aspergillus</i> species

3. Selected food borne diseases

3.1. Cholera

- **Agent:** *Vibrio cholerae*, a gram-negative bacterium
- **Transmission:** Fecal-oral route via contaminated water or food, especially raw seafood and vegetables irrigated with contaminated water
- **Incubation Period:** 2 hours to 5 days
- **Clinical Features:** Profuse watery diarrhea (“rice-water stools”), vomiting, rapid dehydration, hypotension, shock

- **Complications:** Death due to severe dehydration if untreated
- **Prevention:**
 - Safe drinking water and sanitation
 - Proper food handling and cooking
 - Health education in endemic areas
 - Surveillance and outbreak response

Vaccination Details

- **Type:** Oral cholera vaccine (e.g., Dukoral®, Shanchol®)
 - **Route:** Oral
 - **Dose:**
 - Adults and children ≥ 2 years: 2 doses, 1–6 weeks apart
 - **Protection Duration:** Up to 2 years
 - **Indications:**
 - Travelers to endemic areas
 - Populations in outbreak zones
 - Refugee camps and disaster settings
-

3.2. Poliomyelitis (Polio)

- **Agent:** Poliovirus (types 1, 2, and 3), an enterovirus
- **Transmission:** Fecal-oral route via contaminated food, water, or hands

- **Reservoir:** Humans
- **Incubation Period:** 7–14 days
- **Clinical Features:**
 - Most infections are asymptomatic
 - Minor illness: fever, sore throat, headache
 - Major illness: aseptic meningitis, flaccid paralysis, respiratory failure
- **Complications:** Permanent paralysis, post-polio syndrome
- **Prevention:**
 - Hygiene and sanitation
 - Mass immunization campaigns

Vaccination Details

- **Types:**
 - **Oral Polio Vaccine (OPV)** – live attenuated
 - **Inactivated Polio Vaccine (IPV)** – injectable
- **Route:**
 - OPV: Oral drops
 - IPV: Intramuscular or subcutaneous
- **Schedule** (Egypt national program):
 - OPV at birth, 2, 4, and 6 months
 - IPV at 4 and 6 months

- Booster doses at 18 months and 4 years
 - **Indications:**
 - All children under 5
 - Travelers to polio-endemic regions
 - Healthcare workers in outbreak zones
-

3.3. Hepatitis A

Causative Agent:

Hepatitis A virus (HAV), a non-enveloped RNA virus from the Picornaviridae family.

Reservoir:

Humans (no chronic carrier state).

Mode of Transmission:

Fecal-oral route via contaminated food, water, or close personal contact.

Incubation Period:

15–50 days (average 28 days)

Clinical Features:

- Fever, malaise, nausea, abdominal discomfort
- Jaundice, dark urine, pale stools
- Often asymptomatic in children

Diagnosis:

- Detection of anti-HAV IgM antibodies
- Elevated liver enzymes (ALT, AST)

Prevention and Control:

- Hepatitis A vaccine
- Improved sanitation and hygiene
- Safe food and water practices
- Post-exposure prophylaxis with vaccine or immunoglobulin

Epidemiological Notes:

- Endemic in areas with poor sanitation
- Outbreaks common in schools, food service, and refugee settings
- Surveillance essential for outbreak detection

3.4. Typhoid Fever

Causative Agent:

Salmonella enterica serotype Typhi (Gram-negative bacillus)

Reservoir:

Humans (chronic carriers may shed bacteria for years)

Mode of Transmission:

Fecal-oral route via contaminated food or water

Incubation Period:

7–14 days

Clinical Features:

- Sustained high fever, abdominal pain
- Rose-colored spots on trunk
- Constipation or diarrhea, hepatosplenomegaly

Diagnosis:

- Blood culture (gold standard in early phase)
- Stool and urine cultures
- Widal test (limited specificity)

Prevention and Control:

- Typhoid vaccination (Vi polysaccharide or conjugate vaccine)
- Safe water and food hygiene
- Identification and treatment of carriers
- Public health education

Epidemiological Notes:

- Endemic in many low- and middle-income countries
- Risk increases during floods and poor sanitation
- WHO recommends vaccination in high-risk populations

3.5. Brucellosis

Causative Agent:

Brucella species (e.g., *B. melitensis*, *B. abortus*, *B. suis*)

Reservoir:

Domestic animals (goats, sheep, cattle, pigs)

Mode of Transmission:

- Direct contact with infected animals or products
- Ingestion of unpasteurized dairy
- Inhalation in laboratory settings

Incubation Period:

5 days to several weeks

Clinical Features:

- Undulating fever, night sweats
- Arthralgia, fatigue, weight loss
- Hepatosplenomegaly, lymphadenopathy

Diagnosis:

- Blood culture (slow-growing organism)
- Serological tests (e.g., ELISA, SAT)
- PCR in specialized labs

Prevention and Control:

- Pasteurization of milk
- Animal vaccination and control

- Protective measures for high-risk occupations
- Notification and surveillance

Epidemiological Notes:

- Zoonotic disease with occupational relevance
 - Endemic in Mediterranean, Middle East, Latin America
 - Often underreported due to nonspecific symptoms
-

4. General Sources of Contamination

- Infected food handlers
 - Improper cooking or storage
 - Cross-contamination between raw and cooked food
 - Contaminated water used in food preparation
 - Poor sanitation and hygiene practices
-

5. Prevention and Control

Strategy	Examples
Food Safety Practices	Handwashing, clean utensils, safe cooking temperatures
Public Health Measures	Surveillance, outbreak investigation, food inspection

Strategy	Examples
Education and Awareness	Community health campaigns, school programs
Vaccination	Hepatitis A, Typhoid, Cholera, and Polio vaccines for high-risk groups
Regulation and Enforcement	Food labeling, hygiene standards, penalties for violations

End-of-Chapter Assessment

Multiple Choice and True/False Questions

1. True or False: Cholera is primarily transmitted through airborne droplets.

Answer: False

2. Which vaccine is administered orally to prevent polio?

- A. IPV
- B. OPV
- C. TCV
- D. Hepatitis A vaccine

Answer: B

3. **Hepatitis A is commonly transmitted through contaminated blood and body fluids.**

False

4. Typhoid fever can be spread by chronic human carriers who shed *Salmonella Typhi* in their stool.

True

5. Brucellosis is a viral infection primarily spread through airborne droplets in crowded urban settings.

False

Chapter 7: Droplet-Borne Diseases

Intended Learning Objectives

By the end of this chapter, students should be able to:

- 1. Describe the causative agents and transmission routes of major droplet-spread diseases.**
 - 2. Recognize the clinical features and complications of each disease.**
 - 3. Understand the epidemiological patterns and risk groups.**
 - 4. Apply prevention and control strategies, including vaccination and outbreak response.**
 - 5. Interpret surveillance data and contribute to public health interventions.**
-

1. Introduction

Droplet-borne diseases are transmitted through respiratory droplets expelled during coughing, sneezing, talking, or certain medical procedures. These droplets typically travel short distances (<1 meter) and deposit on mucous membranes of nearby individuals. Such diseases are highly contagious and pose significant public health challenges, especially in crowded or healthcare settings.

2. Overview of Transmission

Feature	Droplet Transmission
Particle size	>5 microns
Distance of spread	Usually <1 meter
Mode of entry	Mucous membranes (eyes, nose, mouth)
Environmental survival	Limited
Prevention	Masks, distancing, vaccination, hygiene

3. Key Droplet-Borne Diseases

3.1 Influenza

Causative Agent:

Influenza virus (Types A, B, C); Type A causes pandemics.

Transmission:

Respiratory droplets, contaminated surfaces, close contact.

Incubation Period:

1–4 days

Clinical Features:

- Sudden onset of fever, chills, headache**
- Myalgia, sore throat, dry cough**
- Fatigue, sometimes gastrointestinal symptoms in children**

Complications:

- **Pneumonia (viral or secondary bacterial)**
- **Exacerbation of chronic diseases**
- **Death in high-risk groups**

Prevention:

- **Annual influenza vaccination**
- **Respiratory hygiene and handwashing**
- **Antiviral prophylaxis in outbreaks**

Epidemiological Notes:

- **Seasonal peaks in winter**
 - **High-risk groups: elderly, pregnant women, healthcare workers**
 - **Surveillance through sentinel sites and virological testing**
-

3.2 Measles

Causative Agent:

Measles virus (Paramyxovirus family)

Transmission:

Highly contagious via respiratory droplets and airborne spread

Incubation Period:

7–14 days

Clinical Features:

- **High fever, cough, coryza, conjunctivitis**
- **Koplik spots (white lesions on buccal mucosa)**
- **Maculopapular rash starting on face and spreading downward**

Complications:

- **Otitis media, pneumonia, encephalitis**
- **Subacute sclerosing panencephalitis (rare, late complication)**

Prevention:

- **Measles vaccine (part of MMR)**
- **Isolation of cases**
- **Mass immunization campaigns during outbreaks**

Epidemiological Notes:

- **Endemic in areas with low vaccine coverage**
- **Outbreaks common in refugee camps, schools**
- **WHO targets measles elimination through $\geq 95\%$ vaccine coverage**

3.3 Mumps

Causative Agent:

Mumps virus (Paramyxovirus family)

Transmission:

Respiratory droplets, direct contact with saliva

Incubation Period:

16–18 days

Clinical Features:

- **Fever, headache, malaise**
- **Painful swelling of parotid glands (unilateral or bilateral)**
- **Difficulty chewing or swallowing**

Complications:

- **Orchitis (in post-pubertal males), oophoritis**
- **Meningitis, pancreatitis, deafness**

Prevention:

- **Mumps vaccine (part of MMR)**
- **Isolation of cases for 5 days after onset of parotitis**

Epidemiological Notes:

- **Occurs in unvaccinated populations**
 - **Outbreaks in universities, military camps**
 - **Vaccine coverage reduces incidence significantly**
-

3.4 Rubella

Causative Agent:

Rubella virus (Togavirus family)

Transmission:

Respiratory droplets; vertical transmission during pregnancy

Incubation Period:

14–21 days

Clinical Features:

- **Mild fever, lymphadenopathy (especially postauricular)**
- **Fine pink rash starting on face and spreading**
- **Often asymptomatic in children**

Complications:

- **Congenital Rubella Syndrome (CRS): cataracts, deafness, heart defects**
- **Arthralgia in adults**

Prevention:

- **Rubella vaccine (part of MMR)**
- **Vaccination of women of childbearing age**
- **Surveillance of CRS cases**

Epidemiological Notes:

- **CRS is preventable through immunization**
- **Rubella elimination is a global goal**
- **Serological screening recommended before pregnancy**

3.5 Whooping Cough (Pertussis)

Causative Agent:

***Bordetella pertussis* (Gram-negative coccobacillus)**

Transmission:

Respiratory droplets; highly contagious

Incubation Period:

7–10 days

Clinical Features:

- **Catarrhal phase: mild cough, rhinorrhea**
- **Paroxysmal phase: severe coughing fits with inspiratory “whoop”**
- **Post-tussive vomiting, cyanosis in infants**

Complications:

- **Pneumonia, seizures, encephalopathy**
- **Death in infants <6 months**

Prevention:

- **DTaP vaccine in childhood; Tdap booster in adolescence/adulthood**
- **Cocooning strategy for newborns**
- **Antibiotic prophylaxis for close contacts**

Epidemiological Notes:

- **Resurgence due to waning immunity**
- **Surveillance essential for early detection**
- **Vaccine coverage critical for infant protection**

3.6 Meningococcal Meningitis

Causative Agent:

Neisseria meningitidis (serogroups A, B, C, W, Y)

Transmission:

Respiratory droplets; close contact in crowded settings

Incubation Period:

2–10 days

Clinical Features:

- Sudden fever, headache, neck stiffness
- Photophobia, altered consciousness
- Petechial or purpuric rash (invasive disease)

Complications:

- Septicemia, shock, death
- Neurological sequelae: hearing loss, cognitive impairment

Prevention:

- Meningococcal vaccines (conjugate and polysaccharide)
- Chemoprophylaxis for close contacts (e.g., rifampicin)
- Isolation of cases and outbreak control

Epidemiological Notes:

- Epidemics in “meningitis belt” of sub-Saharan Africa

- Surveillance and mass vaccination campaigns essential
- School and dormitory outbreaks require rapid response

4. Summary Table

Disease	Agent	Incubation	Key Symptoms	Prevention
Influenza	Influenza virus	1–4 days	Fever, cough, myalgia	Annual vaccine
Measles	Measles virus	7–14 days	Fever, rash, Koplik spots	MMR vaccine
Mumps	Mumps virus	16–18 days	Parotid swelling, fever	MMR vaccine
Rubella	Rubella virus	14–21 days	Rash, lymphadenopathy	MMR vaccine
Pertussis	<i>B. pertussis</i>	7–10 days	Whooping cough, vomiting	DTaP/Tdap vaccine
Meningitis	<i>N. meningitidis</i>	2–10 days	Fever, neck stiffness, rash	Meningococcal vaccine

1. Measles can be transmitted through airborne particles as well as respiratory droplets.

True

2. Rubella infection during pregnancy poses no risk to the fetus.

False

3. Mumps primarily affects the liver and causes jaundice.

False

4. Annual vaccination is recommended to prevent seasonal influenza.

True

5. Pertussis is most dangerous in infants under six months of age.

True

Chapter 8: Blood-Borne Diseases

Intended Learning Objectives

By the end of this chapter, students should be able to:

- 1. Define blood-borne diseases and explain their transmission routes.**
 - 2. Describe the epidemiology, clinical progression, and complications of HBV, HCV, and HIV.**
 - 3. Identify high-risk populations and occupational exposure risks.**
 - 4. Apply prevention strategies including vaccination, screening, and post-exposure prophylaxis.**
 - 5. Interpret surveillance data and contribute to public health control programs.**
-

1. Introduction

Blood-borne diseases are infections transmitted through exposure to infected blood or body fluids. They represent a major public health concern due to their potential for chronic illness, cancer, and death. Healthcare workers, intravenous drug users, and recipients of unscreened blood products are particularly vulnerable. The three most significant blood-borne pathogens are:

- Hepatitis B virus (HBV)**
- Hepatitis C virus (HCV)**

- **Human Immunodeficiency Virus (HIV)**

These diseases share common transmission routes but differ in their natural history, treatment options, and prevention strategies.

2. Modes of Transmission

Exposure Route	Examples
Percutaneous	Needle-stick injuries, cuts with contaminated sharps
Mucosal	Splash to eyes, nose, or mouth with infected fluids
Sexual contact	Unprotected intercourse, especially with multiple partners
Vertical transmission	From mother to child during birth or breastfeeding
Blood transfusion	Especially in settings with poor screening protocols
Shared injection equipment	Among intravenous drug users
Unsafe medical/dental practices	Reuse of needles, poor sterilization

3. Hepatitis B Virus (HBV)

3.1 Causative Agent

HBV is a double-stranded DNA virus from the Hepadnaviridae family. It is highly infectious—100 times more than HIV—and can survive outside the body for up to 7 days.

3.2 Epidemiology

- **Global burden: ~296 million chronic carriers (WHO, 2023)**
- **Endemic in Africa, Asia, and parts of the Middle East**
- **Egypt: Intermediate endemicity; universal infant vaccination introduced in 1992**

3.3 Incubation Period

30–180 days (average 60–90 days)

3.4 Clinical Features

- **Acute phase: Fever, malaise, nausea, jaundice**
- **Chronic phase: Often asymptomatic; may progress to cirrhosis or hepatocellular carcinoma**
- **Fulminant hepatitis: Rare but life-threatening**

3.5 Diagnosis

- **HBsAg: Indicates current infection**
- **Anti-HBs: Indicates immunity (natural or vaccine-induced)**
- **HBeAg and HBV DNA: Assess infectivity and viral replication**
- **Liver function tests: ALT, AST elevation**

3.6 Prevention

- **Vaccination: 3-dose schedule; highly effective**
- **Post-exposure prophylaxis: HBIG + vaccine within 24 hours**
- **Safe practices: Blood screening, sterile equipment, education**

3.7 Public Health Notes

- **HBV vaccine is part of Egypt's Expanded Program on Immunization (EPI)**
 - **Screening of pregnant women prevents vertical transmission**
 - **Chronic carriers require regular monitoring and antiviral therapy**
-

4. Hepatitis C Virus (HCV)

4.1 Causative Agent

HCV is a single-stranded RNA virus from the Flaviviridae family. It has multiple genotypes; genotype 4 is most common in Egypt.

4.2 Epidemiology

- **Global burden: ~58 million chronic cases (WHO, 2023)**
- **Egypt had one of the highest prevalence rates globally due to past schistosomiasis campaigns**
- **National screening and treatment programs have significantly reduced incidence**

4.3 Incubation Period

2 weeks to 6 months

4.4 Clinical Features

- **Acute phase: Often asymptomatic or mild flu-like symptoms**
- **Chronic phase: Fatigue, liver enzyme elevation**

- **Complications: Cirrhosis, hepatocellular carcinoma**

4.5 Diagnosis

- **Anti-HCV antibodies: Initial screening**
- **HCV RNA (PCR): Confirms active infection**
- **Genotyping: Guides treatment**
- **Fibroscan or biopsy: Assesses liver damage**

4.6 Prevention

- **No vaccine available**
- **Blood safety: Rigorous screening of donors**
- **Harm reduction: Needle exchange programs, education**
- **Infection control: Sterile medical procedures**

4.7 Public Health Notes

- **Egypt's 100 Million Health Initiative screened millions for HCV**
- **Direct-acting antivirals (DAAs) offer >95% cure rate**
- **WHO aims to eliminate HCV as a public health threat by 2030**

5. Human Immunodeficiency Virus (HIV/AIDS)

5.1 Causative Agent

HIV is a retrovirus that targets CD4⁺ T lymphocytes, leading to progressive immune deficiency.

5.2 Epidemiology

- **Global burden: ~39 million people living with HIV (UNAIDS, 2023)**
- **Egypt: Low prevalence (<0.1%), but increasing among key populations**
- **High-risk groups: MSM, sex workers, IV drug users**

5.3 Incubation Period

Weeks to years; AIDS may develop 5–10 years after infection

5.4 Clinical Features

- **Acute phase: Fever, rash, lymphadenopathy**
- **Chronic phase: Asymptomatic or mild symptoms**
- **AIDS: Opportunistic infections (TB, PCP), malignancies (Kaposi's sarcoma), wasting**

5.5 Diagnosis

- **Rapid test or ELISA: Screening**
- **Western blot or PCR: Confirmation**
- **CD4 count and viral load: Monitor disease progression and treatment response**

5.6 Prevention

- **No vaccine available**
- **Safe sex practices: Condom use, education**
- **PEP: Antiretroviral therapy within 2 hours of exposure**
- **PrEP: Daily antiretrovirals for high-risk individuals**

- **Blood safety and sterile equipment**

5.7 Public Health Notes

- **UNAIDS 95-95-95 targets: 95% diagnosed, 95% on treatment, 95% virally suppressed**
 - **Stigma remains a barrier to care**
 - **Integration with TB and reproductive health services improves outcomes**
-

6. Summary Table

Disease	Agent	Incubation	Chronic Risk	Vaccine	Key Prevention	Treatment
HBV	DNA virus	30–180 days	Yes	Yes	Vaccine, HBIG	Antivirals
HCV	RNA virus	2–24 weeks	Yes	No	Blood safety	DAAs
HIV	RNA virus	Variable	Yes	No	Safe sex, PEP	ART

True/False Questions

1. **Hepatitis B is more infectious than HIV and can survive outside the body for several days.**
True
2. **There is an effective vaccine available for Hepatitis C.**
False

3. HIV infection always presents with symptoms in the early phase.

False

4. Post-exposure prophylaxis for HIV should be started within 72 hours of exposure.

True

5. HCV infection can be cured with direct-acting antiviral therapy.

True

Chapter 9: Contact-Borne Diseases

Intended Learning Objectives

By the end of this chapter, students should be able to:

- 1. Define contact transmission and distinguish between direct and indirect contact.**
 - 2. Describe the epidemiology, clinical features, and prevention of key contact-borne diseases.**
 - 3. Explain the pathogenesis and control strategies for rabies and tetanus.**
 - 4. Understand vaccination schedules for rabies and tetanus, including protocols during pregnancy.**
 - 5. Apply public health measures to reduce transmission in healthcare and community settings.**
-

1. Introduction

Contact-borne diseases are transmitted through physical contact with infected individuals, contaminated surfaces, or animal bites. Transmission may be:

- Direct contact: skin-to-skin, mucous membranes, bites, scratches**
- Indirect contact: via contaminated objects (fomites), such as towels, instruments, or soil**

These diseases range from mild skin infections to life-threatening neuroinvasive conditions. Rabies and tetanus are two critical contact-transmitted diseases with high fatality rates if untreated, yet both are vaccine-preventable.

2. Overview of Transmission

Type	Mechanism	Examples
Direct contact	Skin, mucosa, bites, scratches	Rabies, scabies, impetigo
Indirect contact	Contaminated surfaces, soil, instruments	Tetanus, MRSA, ringworm

3. Key Contact-Borne Diseases

3.1 Rabies

Causative Agent:

Rabies virus (*Lyssavirus*, Rhabdoviridae family)

Reservoirs:

Dogs (primary in Egypt), bats, foxes, raccoons

Mode of Transmission:

- Bite or scratch from infected animal

- **Saliva contact with broken skin or mucosa**

Incubation Period:

Typically 1–3 months (can range from days to years depending on bite site and viral load)

Clinical Features:

- **Prodromal phase: Fever, malaise, pain at bite site**
- **Neurological phase: Hydrophobia, agitation, hallucinations, paralysis**
- **Coma and death: Once symptoms appear, rabies is almost always fatal**

Diagnosis:

- **Clinical suspicion based on exposure**
- **RT-PCR from saliva, skin biopsy, or CSF**
- **Postmortem confirmation from brain tissue**

Prevention and Control:

- **Pre-exposure prophylaxis (PrEP): For high-risk groups (veterinarians, lab workers)**
 - **Post-exposure prophylaxis (PEP): Immediate wound cleaning + vaccine ± rabies immunoglobulin (RIG)**
-



Rabies Vaccination Schedule

- ◆ **Pre-Exposure Prophylaxis (PrEP)**
- **Indicated for: High-risk occupations**

- **Schedule:**
 - **Day 0, Day 7, Day 21 or 28 (intramuscular or intradermal)**
 - **Booster every 3–5 years if risk persists**
- ◆ **Post-Exposure Prophylaxis (PEP)**
- **For previously unvaccinated individuals:**
 - **Rabies vaccine: 4 doses on Days 0, 3, 7, and 14**
 - **Rabies Immunoglobulin (RIG): Infiltrated around wound on Day 0**
- **For previously vaccinated individuals:**
 - **Rabies vaccine only: 2 doses on Days 0 and 3**
 - **No RIG needed**

Public Health Notes:

- **Rabies is a notifiable disease**
 - **Dog vaccination campaigns are essential**
 - **Education on wound care and prompt reporting is critical**
-

3.2 Tetanus

Causative Agent:

***Clostridium tetani* – anaerobic, spore-forming bacterium**

Reservoir:

Soil, dust, animal feces

Mode of Transmission:

- **Entry of spores into wounds, burns, or punctures**
- **Non-communicable: not spread person-to-person**

Incubation Period:

3–21 days (average 7 days)

Clinical Features:

- **Early signs: Jaw stiffness (trismus), neck rigidity**
- **Progression: Muscle spasms, opisthotonos, respiratory failure**
- **Neonatal tetanus: Infection via contaminated umbilical stump**

Diagnosis:

- **Clinical presentation**
- **No specific laboratory test; diagnosis is clinical**

Prevention and Control:

- **Vaccination is the cornerstone**
- **Proper wound care and hygiene**
- **Tetanus toxoid boosters for injuries**



Tetanus Vaccination Schedule

- ◆ **Routine Childhood Immunization (Egypt EPI)**
- **DTaP vaccine at:**

- **2 months**
- **4 months**
- **6 months**
- **Booster at 18 months**
- **Booster at 4–6 years**
- **Tdap booster at 11–12 years**

◆ **Adult Boosters**

- **Td vaccine every 10 years**
- **Tdap once in adulthood if not previously received**

◆ **Post-Exposure Prophylaxis**

Wound Type	Vaccination History	Recommended Action
Clean minor	≥3 doses	No vaccine needed
Dirty wound	≥3 doses	Td booster if >5 years
Any wound	<3 doses or unknown	Td + TIG (Tetanus Immunoglobulin)

Tetanus Vaccination in Pregnancy

Goal: Prevent maternal and neonatal tetanus

Recommended Schedule (WHO & Egyptian MOH):

Dose	Timing	Protection
TT1	After first trimester	No protection yet
TT2	≥4 weeks after TT1	3 years protection
TT3	≥6 months after TT2	5 years protection
TT4	≥1 year after TT3	10 years protection
TT5	≥1 year after TT4	Lifetime protection

4. Other Contact-Borne Diseases (Brief Overview)

Disease	Agent	Transmission	Prevention
Impetigo	<i>S. aureus, S. pyogenes</i>	Skin contact	Hygiene, topical antibiotics
Scabies	<i>Sarcoptes scabiei</i>	Prolonged skin contact	Treat contacts, wash linens
Ringworm	Dermatophytes	Skin/fomite contact	Antifungals, hygiene
MRSA	Resistant <i>S. aureus</i>	Direct/indirect contact	Contact precautions

Multiple Choice Questions (MCQs)

1. Which of the following diseases is non-communicable and acquired through contaminated wounds?

- A. Rabies**
- B. Tetanus**
- C. Scabies**
- D. MRSA**

Correct Answer: B. Tetanus

2. Rabies post-exposure prophylaxis includes:

- A. Antibiotics and tetanus vaccine**
- B. Rabies vaccine and immunoglobulin**
- C. Antifungal cream and wound dressing**
- D. Isolation and antiviral tablets**

Correct Answer: B. Rabies vaccine and immunoglobulin

3. How often should adults receive a tetanus booster if fully vaccinated?

- A. Every 5 years**
- B. Every 10 years**
- C. Once in a lifetime**
- D. Every year**

Correct Answer: B. Every 10 years

4. Which vaccine is recommended during pregnancy to prevent neonatal tetanus?

- A. DTaP**

B. Rabies vaccine

C. TT (Tetanus Toxoid)

D. Hepatitis B vaccine

Correct Answer: C. TT (Tetanus Toxoid)

5. Which of the following statements about rabies is TRUE?

A. It is curable after symptoms appear

B. It is transmitted through contaminated water

C. It is preventable by vaccination and wound care

D. It is a fungal disease

Correct Answer: C. It is preventable by vaccination and wound care

Chapter 10: Epidemiology of Non-Communicable Diseases (NCDs)

Intended Learning Objectives

By the end of this chapter, students should be able to:

- 1. Define non-communicable diseases and distinguish them from infectious diseases.**
 - 2. Describe the global and national burden of NCDs.**
 - 3. Identify major NCDs and their risk factors.**
 - 4. Explain the components of the Healthy Eating Plate and its role in disease prevention.**
 - 5. Recommend evidence-based dietary and physical activity guidelines.**
 - 6. Apply prevention strategies at individual and population levels.**
-

1. Introduction

Non-communicable diseases (NCDs) are chronic conditions that are not spread from person to person. They develop over time, often silently, and are influenced by lifestyle, environment, and genetics. NCDs are the leading cause of death worldwide, responsible for over 70% of global mortality, and are increasingly prevalent in Egypt due to urbanization, dietary changes, and reduced physical activity.

2. Major Non-Communicable Diseases

Disease Group	Examples
Cardiovascular diseases	Hypertension, coronary artery disease, stroke
Cancers	Breast, lung, colorectal, cervical
Chronic respiratory diseases	Asthma, COPD
Diabetes mellitus	Type 1 and Type 2
Mental health disorders	Depression, anxiety, schizophrenia
Musculoskeletal disorders	Osteoarthritis, osteoporosis

3. Risk Factors

Modifiable Risk Factors

- **Tobacco use**
- **Unhealthy diet**
- **Physical inactivity**
- **Excessive alcohol consumption**
- **Obesity**
- **High blood pressure, high cholesterol, high blood sugar**

Non-Modifiable Risk Factors

- Age
 - Gender
 - Genetic predisposition
 - Family history
-

4. The Healthy Eating Plate

The Healthy Eating Plate is a visual guide developed by nutrition experts to promote balanced, disease-preventing meals. It replaces outdated food pyramids with a clear, practical model.

Components of the Healthy Eating Plate

Section	Recommendations
Half of the plate	Vegetables and fruits – variety and color; limit potatoes and starchy vegetables
One quarter	Whole grains – brown rice, whole wheat, oats; avoid refined grains
One quarter	Healthy proteins – fish, poultry, legumes, nuts; limit red and processed meats
Healthy oils	Use olive, canola, sunflower oils; avoid trans fats and hydrogenated oils

Section	Recommendations
Water	Preferred drink; limit sugary beverages and fruit juices
Physical activity	Shown beside the plate to emphasize daily movement

5. Recommended Healthy Diet

Increase Consumption Of:

- Fruits and vegetables: ≥ 5 servings/day
- Whole grains: Brown rice, whole wheat bread, bulgur
- Legumes: Lentils, beans, chickpeas
- Healthy fats: Olive oil, nuts, seeds
- Low-fat dairy: Yogurt, laban, cheese

Limit Consumption Of:

- Added sugars: $<10\%$ of daily energy
 - Salt: $<5\text{g/day}$ (≈ 1 teaspoon)
 - Saturated and trans fats: Found in fried foods, margarine, pastries
 - Processed meats: Sausages, luncheon meats
-

6. Recommended Physical Activity

WHO Guidelines for Adults (18–64 years)

- **150–300 minutes/week of moderate-intensity aerobic activity**
- **OR 75–150 minutes/week of vigorous-intensity activity**
- **Muscle-strengthening activities ≥ 2 days/week**
- **Reduce sedentary time; break long sitting periods**

Children and Adolescents (5–17 years)

- **≥ 60 minutes/day of moderate to vigorous activity**
- **Include aerobic and bone-strengthening exercises**

Older Adults (65+ years)

- **Same as adults, with added focus on balance and flexibility**
- **Activities should be adapted to functional ability**

7. Prevention and Control Strategies

Individual-Level

- **Lifestyle modification: diet, exercise, smoking cessation**
- **Routine screening: blood pressure, glucose, BMI**
- **Medication adherence and chronic disease management**

Population-Level

Strategy	Examples
Health education	School and media campaigns
Policy and regulation	Tobacco taxation, food labeling, salt reduction
Environmental changes	Walkable cities, clean air initiatives
Health system strengthening	NCD clinics, universal coverage, integrated care

✔ Multiple Choice Questions (MCQs)

1. Which component should occupy half of the Healthy Eating Plate?

- A. Whole grains**
- B. Fruits and vegetables**
- C. Proteins**
- D. Dairy products**

Correct Answer: B. Fruits and vegetables

2. Which of the following is a recommended source of healthy fat?

- A. Margarine**
- B. Butter**
- C. Olive oil**
- D. Palm oil**

Correct Answer: C. Olive oil

3. WHO recommends how many minutes of moderate-intensity physical activity per week for adults?

- A. 60 minutes**
- B. 150–300 minutes**
- C. 500 minutes**
- D. 30 minutes only**

Correct Answer: B. 150–300 minutes

4. Which of the following is a modifiable risk factor for NCDs?

- A. Age**
- B. Gender**
- C. Family history**
- D. Physical inactivity**

Correct Answer: D. Physical inactivity

5. Which dietary pattern is most associated with cardiovascular protection?

- A. High-fat ketogenic diet**
- B. Mediterranean diet**
- C. Western fast-food diet**
- D. Low-protein diet**

Correct Answer: B. Mediterranean diet

Chapter 11: Measurements in epidemiology

objectives

- Identify basic tools in epidemiology.
- Identify the difference between ratio, proportion and rate
- Recognize the difference between incidence and prevalence.
- Recognize the difference between primary and secondary attack rate
- Recognize the commonly used mortality rates
- Calculate ratio, proportion, incidence, prevalence and crude death rate, case fatality rate, infant and maternal mortality rate.

The magnitude of the health problem in the community is measured in terms of diseases (morbidity) and deaths (mortality). It is expressed in terms of rates, ratios and proportions, which are called 'Basic tools' in epidemiology.

Ratio

- This expresses a relation between the two quantities. The numerator is not a component of the denominator. It is expressed as x/y or $x : y$.
- Example, out of 100 patients attending the dental clinic, 25 were males and 75 were females.
- So, male/female ratio= $25/75=1/3$ or 1:3

Proportion

- This also expresses relation between two quantities, but here the numerator is always included in the denominator. It is expressed as percentage.
- In the previous example, male proportion = males/total
- $=25/100 \times 100 = 25\%$

Rate

- It measures the occurrence of an event (disease or death) in a given population, during a given period of time, usually one year, e.g. death rate, birth rate, etc. Usually it is expressed per 1000 mid year population (MYP).
- Example, death rate is the rate at which the people are dying in a given area, during a given period of time, per 1000 MYP.

1. Measurement of morbidity

- This means determining the quantum of disease in the community (i.e. disease load).
- morbidity is measured and expressed in two ways
 - A. Incidence rate.
 - B. Prevalence rate.

A. Incidence rate

It is the occurrence of only NEW cases of a specified disease in a given area, during a given period of time (minimum one year). It is expressed for 1000 population at risk.

No. of new cases of in a given area and period

$$\text{Incidence rate} = \frac{\text{No. of new cases of in a given area and period}}{\text{Population at risk during that period}} \times 1,000$$

For example, if 200 new cases of tuberculosis occur in a population of 1,00,000 in a year.

The incidence rate would be: $IR = (200 / 1,00,000) \times 1,000 = 2$ per 1,000 population at risk

The incidence rate in epidemic is called attack rate and is expressed in percentage not per 1000 population.

Ex, Of 100 persons who attended a birthday party, 20 subsequently developed gastroenteritis. The attack rate of gastroenteritis = $(20/100) \times 100 = 20\%$

B. Prevalence

- It refers to the existence of all cases (both old and new) of a specified disease in a given population.

No. of all current cases (Old + New)

$$\text{Prevalence rate} = \frac{\text{No. of all current cases (Old + New)}}{\text{Total population}} \times 1000$$

- at a given point (hour, day or week) -----point prevalence
- or a given period of time (months or years) ----period prevalence

The prevalence and incidence of disease are frequently confused. They are similar but differ in what cases are included in the numerator. Numerator of Incidence = new cases occurring during a given time period Numerator of Prevalence = all cases present during a given time period. As you can see, the numerator of an incidence rate consists only of persons whose illness began during a specified interval. The numerator for prevalence includes all persons ill from a specified cause during a specified interval (or at a specified point in time) regardless of when the illness began. It includes not only new cases, but also old cases representing persons who remained ill during some portion of the specified interval. A case is counted in prevalence until death or recovery occurs.

Differences between incidence and prevalence

<i>Incidence</i>	<i>prevalence</i>
Refers to new cases only	Refers to old and new cases
Estimation requires exact knowledge of time of onset of disease.	Such knowledge is not required
Refers to acute cases having short incubation period.	Refers to chronic cases having long incubation period
Related terms are attack rate	Related terms are point prevalence and period prevalence

Incidence rate helps to evaluate the control measures	Prevalence rate helps for health planning
--	--

Attack rate

An attack rate is a variant of an incidence rate, applied to a narrowly defined population observed for a limited time, such as during an epidemic. The attack rate is usually expressed as a percent. For a defined population (the population at risk), during a limited time period, $\text{Attack rate} = \frac{\text{Number of new cases among the population during the period}}{\text{Population at risk at the beginning of the period}} \times 100$

Example

Of 75 persons who attended a church picnic, 46 subsequently developed gastroenteritis.

To calculate the attack rate of gastroenteritis we first define the numerator and denominator:

$x = \text{Cases of gastroenteritis occurring within the incubation period for gastroenteritis among persons who attended the picnic} = 46$

$y = \text{Number of persons at the picnic} = 75$

Then, the attack rate for gastroenteritis is $46/75 \times 100 = 61\%$

Secondary Attack Rate

A secondary attack rate is a measure of the frequency of new cases of a disease among the contacts of known cases. The formula is as follows:

Number of cases among contacts of primary cases during the period

$$\blacksquare \text{ Secondary attack rate} = \frac{\text{-----}}{\text{total number of contacts}} \times 100$$

To calculate the total number of household contacts, we usually subtract the number of primary cases from the total number of people residing in those households.

Example

Seven cases of hepatitis A occurred among 70 children attending a child care center. Each infected child came from a different family. The total number of persons in the 7 affected families was 32. One incubation period later, 5 family members of the 7 infected children also developed hepatitis A. We will calculate the attack rate in the child care center and the secondary attack rate among family contacts of those cases.

1. Attack rate in child care center:

x = cases of hepatitis A among children in child care center = 7

y = number of children enrolled in the child care center = 70

Attack rate = $x/y \times 100 = 7/70 \times 100 = 10\%$

2. Secondary attack rate:

x = cases of hepatitis A among family contacts of children with hepatitis A

= 5

y = number of persons at risk in the families (total number of family members—children already infected) = 32 – 7 = 25

Secondary attack rate = $x/y \times 100 = 5/25 \times 100 = 20\%$

2. Measurement of mortality

The commonly used indicators are:

- Crude death rate
- Cause-specific death rate
- Case fatality rate
- Proportional mortality rate
- Infant mortality rate
- Maternal mortality rate

Crude death rate

No. of deaths from all causes during a given year

CDR=-----x1000

Mid-year population

- Higher the death rate, it means poorer is the health status of the population. Thus, it is an indirect way of assessing the health status of a country.
- It is called ‘Crude’ because it does not specify the age group of deaths, or sex groups of deaths nor the cause of deaths.

Cause-specific death rate

The cause-specific mortality rate is the mortality rate from a specified cause for a population. The numerator is the number of deaths attributed to a specific cause. The denominator remains the size of the population at the midpoint of the time period. For 10n , we use 100,000.

Case-fatality rate

The case-fatality rate is the proportion of persons with a particular condition (cases) who die from that condition. The formula is:

$$\text{Case-fatality} = \frac{\text{Number of cause -specific deaths among the incident cases}}{\text{Number of incident cases}} \times 10n$$

Proportionate mortality

Proportionate mortality describes the proportion of deaths in a specified population over a period of time attributable to different causes. Each cause is expressed as a percentage of all deaths, and the sum of the causes must add to 100%. These proportions are not mortality rates, since the denominator is all deaths, not the population in which the deaths occurred. For a specified population over a specified period,

$$\text{Proportionate mortality} = \frac{\text{Deaths due to a particular cause}}{\text{Deaths from all causes}} \times 100$$

Infant mortality rate (IMR)

The infant mortality rate is one of the most commonly used measures for comparing health services among nations. IMR is an, universally accepted indicator of health status and is usually expressed per 1,000 live births.

number of deaths among children under 1 year of age reported during a
given time period

$$\text{IMR} = \frac{\text{number of deaths among children under 1 year of age reported during a given time period}}{\text{the number of live births reported during the same time period}} \times 100$$

the number of live births reported during the same time period.

Maternal mortality rate (MMR)

The maternal mortality rate is really a ratio used to measure mortality associated with pregnancy. Because maternal mortality is much less common than infant mortality, the maternal mortality rate is usually expressed per 100,000 live births.

the number of deaths assigned to causes related to pregnancy, labour and
puerperium during a given time period

$$\text{MMR} = \frac{\text{the number of deaths assigned to causes related to pregnancy, labour and puerperium during a given time period}}{\text{the number of live births reported during the same time period}} \times 100$$

the number of live births reported during the same time period.

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Chapter 12: Epidemiological studies

Objectives

- Recognize the difference between descriptive and analytic epidemiological studies.
- Identify the main descriptive studies
- Identify the types of analytic studies
- Identify the characteristics and main differences between case control, cohort and cross-sectional studies.

The selection of a research strategy is the core of a research design and is probably the single most important decision the investigator has to make.

Descriptive epidemiology describes pattern of health event / disease occurrence in relation to variables such as (Person, Place, Time). Formulating hypothesis is the main aim of descriptive epidemiological studies.

Main type of descriptive studies are

a. Case reports and case series:

- **Case report** → describes the experience of one case (single patient)
- **Case series** → refers to group of similar cases that describe characteristics of a number of patients with a given disease.

b. Community diagnosis or needs assessment

This kind of study entails collection of data on existing health problems, programmes, achievements, constraints, social stratification, leadership patterns, focal points of resistance or high prevalence, or groups at highest risk. Its purpose is

to identify existing needs and to provide baseline data for the design of further studies or action.

c. Epidemiological description of disease occurrence

This common use of the descriptive approach entails the collection of data on the occurrence and distribution of disease in populations according to specific characteristics of individuals (e.g. age, sex, education, smoking habits, religion, occupation, social class, marital status, health status, personality), place (rural/urban, local, subnational, national, international) and time (epidemic, seasonal, cyclic, secular). A description may also be given of familial characteristics such as birth order, parity, family size, maternal age, birth interval or family type.

d. . Descriptive cross sectional studies (prevalence study or survey)

Cross-sectional studies entail the collection of data on a cross-section of the population, which may comprise the whole population or a proportion (sample) of it.

They provide a prevalence rate at a particular point in time (point prevalence) or over a period of time (period prevalence).

Analytical strategies in epidemiology

- Analytic epidemiology is concerned with the search for causes and effects, or the why and the how.
- We use analytic epidemiology to quantify the association between exposures and outcomes
- Analytic epidemiological studies fall into two categories: experimental and observational.

- In an experimental study, we determine the exposure status for each individual (clinical trial) or community (community trial); we then follow the individuals or communities to detect the effects of the exposure.
- In an observational study, we do not control the exposure status, we simply observe the exposure and outcome status of each study participant.
- observational studies aim at testing hypothesis and experimental studies aim at confirming hypothesis.

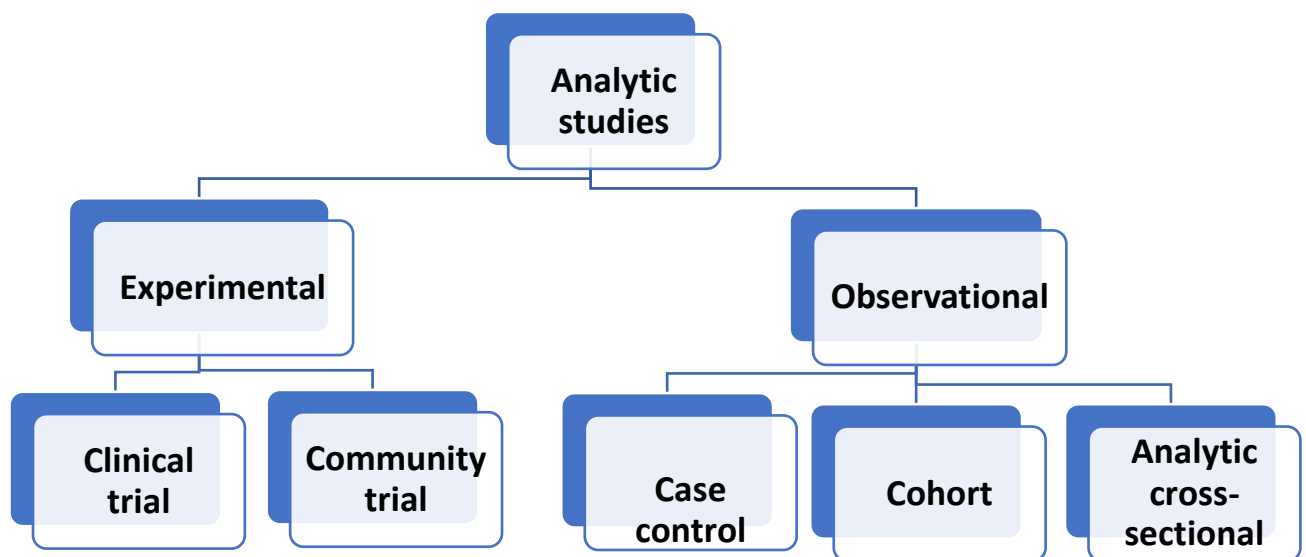


Fig. (10): Analytic studies

Case control studies

The simplest and most commonly used analytical strategy in epidemiology involves the case-control study. It is designed primarily to establish the causes of diseases by investigating associations between exposure to a risk factor and the occurrence of disease. The design is relatively simple, except that it is backward-looking (retrospective) based on the exposure histories of cases and controls. With this type of study, one investigates an association by contrasting the exposure of a series of cases of the specified disease with the exposure pattern of carefully selected control groups free from that particular disease (fig. 11).

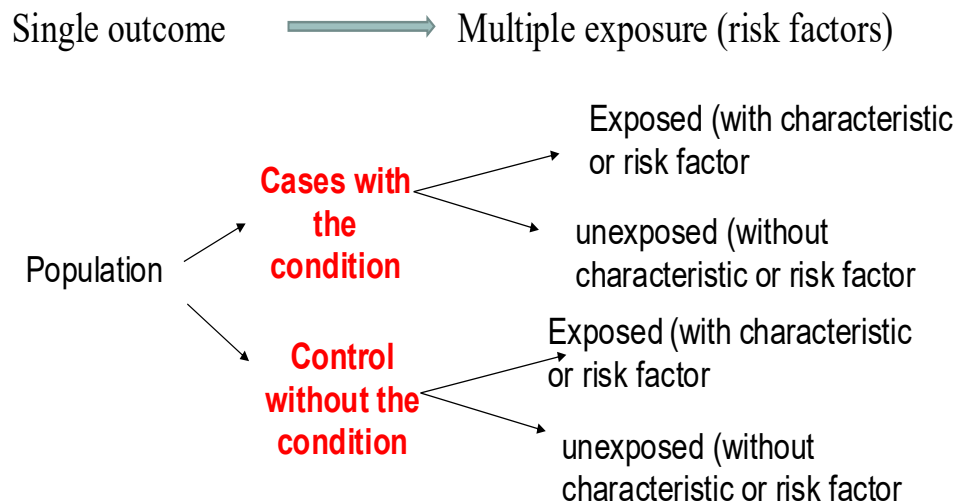
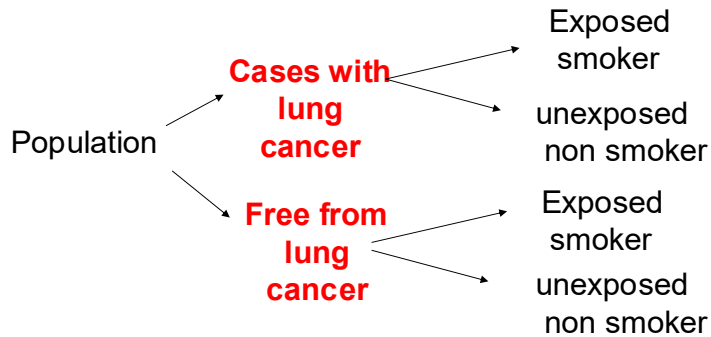


Fig. (11): case control studies

Case control studies

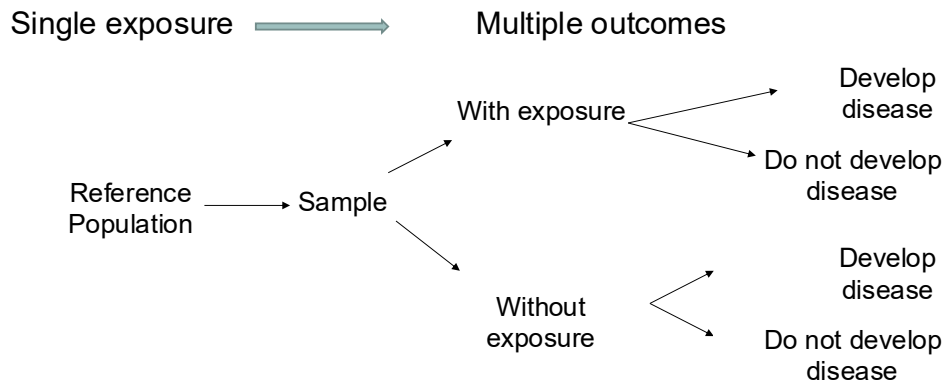
e.g.:



Prospective Cohort studies

The common strategy of cohort studies is to start with a reference population (or a representative sample thereof), some of whom have certain characteristics or attributes relevant to the study (exposed group), with others who do not have those characteristics (unexposed group). Both groups should, at the outset of the study, be free from the condition or conditions under consideration. Both groups are then observed over a specified period to find out the risk each group has of developing the condition(s) of interest. This is illustrated diagrammatically in Figure (12).

Cohort studies



Cohort studies

e.g.:

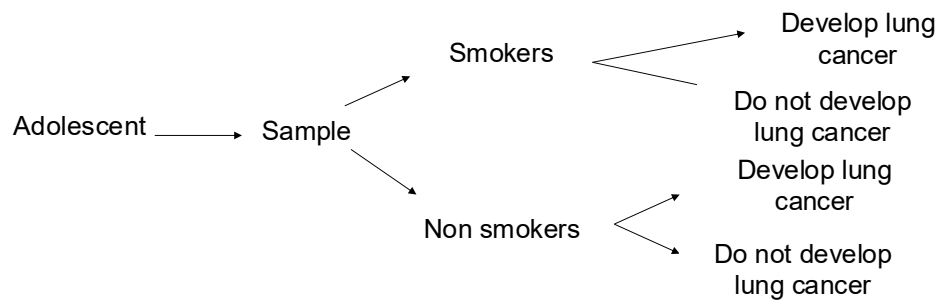


Fig. (12): cohort studies

Differences between case control and cohort studies

Case control	cohort
Less expensive	More expensive
Short time	Long time
Does not need large sample	Need large sample
Can not calculate incidence	Can accurately calculate incidence
Depends on memory	not depend on memory



Analytic cross sectional studies

- When exposure and disease status are assessed simultaneously among individuals in a well-defined population (whole population or population sample). This is illustrated diagrammatically in Figure (13).

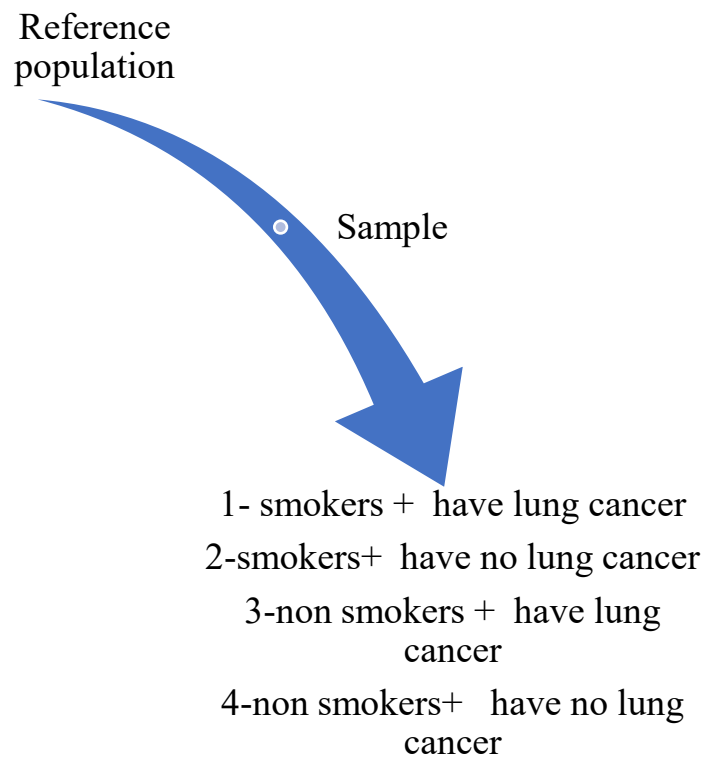
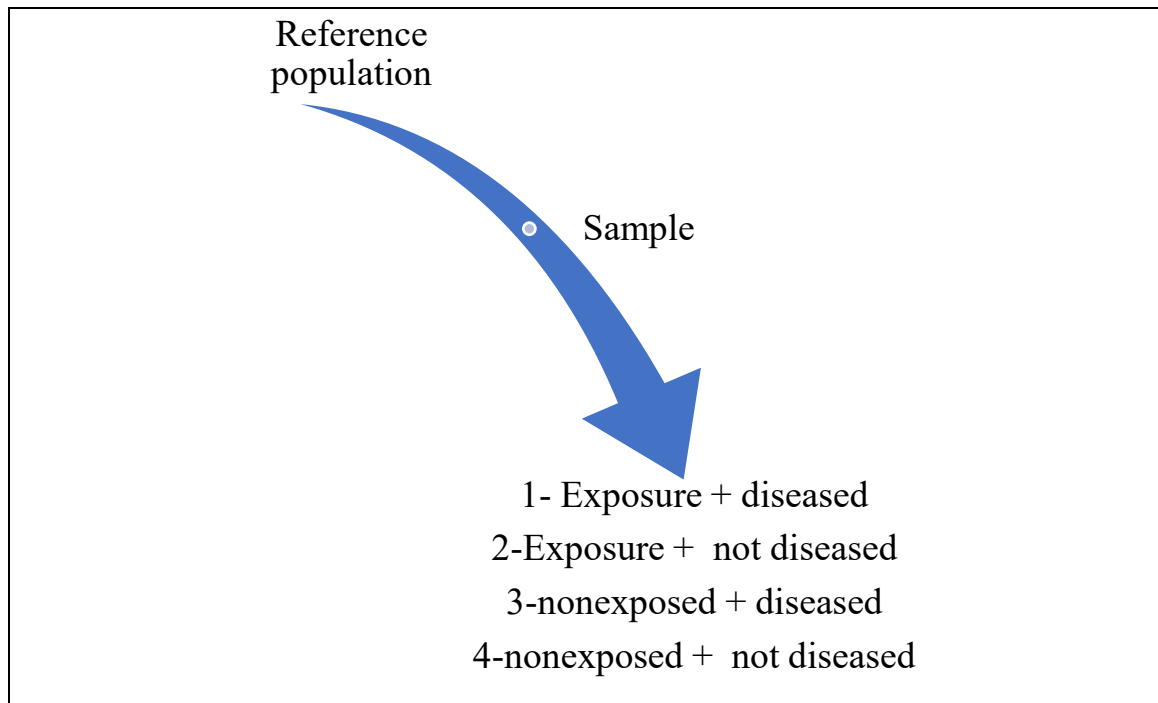


Fig. (13): Analytic cross-sectional studies

Chapter 13: Sampling methods

Objectives

- Identify the definition and aim of sampling
- Recognize the sampling process.
- Recognize the difference between different types of sampling.

Most research studies involve the observation of a sample from some predefined population of interest. In epidemiological studies, for example, a sample of people is observed for exposure to various risk factors, health outcomes and other related variables. The conclusions drawn from the study are often based on generalizing the results observed in the sample to the entire population from which the sample was drawn. Therefore, the accuracy of the conclusions will depend on how well the samples have been collected, and especially on how representative the sample is of the population.

Sampling is a process of choosing a section of the population for observation and study. samples are chosen rather than the whole population for study to minimize the costs, resources, time and manpower.

Process of sampling

1. Before a sample is drawn, the population has to be clearly defined.

- In a population survey, this requires having a list of all the individuals in the population (sampling frame).
- In experiments (such as clinical trials) **this list may not be explicit** at the beginning of the study. For example, a list of inclusion and exclusion criteria

would be specified at the beginning of the trial, defining the general framework for the population.

2. Size of the sample.

Calculated by special equations for each study design. It Can be automatically calculated by using statistical programs (e.g. Epi info)

3. Type the sample

There are two types of sampling

1. Purposive (selective or non-probability sample):

Samples chosen according to person's own judgment. Can't be considered as representative samples ex, (Snow ball sampling which is used when the health problem under studying is hard to reach. e.g.: HIV/AIDS among commercial sex workers or IV drug addicts, homeless individuals)

2. Random (probability sample):

Each unit of population has an equal chance or probability of being chosen. Example; Simple random sample, Systematic random sample, Stratified random sample, cluster sampling and Multistage random sample.

***a.* Simple random sample:**

This is the most common and the simplest of the sampling methods and used for homogenous population. One may use a random number table, or use techniques such as putting the names of the people into a hat and selecting the appropriate number of names blindly. Recently, computer programs have been developed to draw simple random samples from a given population.

***b.* Systemic random sample:**

After a population are arranged in list or form → sample taken as the following examples: e.g. if sample 10 from 200 → initial selection → 10,20,30 ...etc.

***c.* Stratified random sample:**

Used in non homogenous population . In this condition → we divide the population into subgroup (strata) and from each stratum → we take a sample randomly after determining its size.

***d.* Cluster sampling:**

In many administrative surveys, studies are done on large populations which may be geographically quite dispersed. In such cases, clusters may be identified (e.g. households) and random samples of clusters will be included in the study; then every member of the cluster will also be part of the study.

***e.* Multi-stage sampling:**

Many studies, especially large nationwide surveys, will incorporate different sampling methods for different groups, and may be done in several stages.

Exercise

1. 100 population were listed and the estimated sample was 20 subject chosen as follow, 5,10, 15, 20,25,... ect. This sampling technique is
 - Simple random sampling
 - Stratified random sampling
 - Systematic random sampling
 - Cluster sampling

2. The population should not be homogenous in simple random sampling.
 - True
 - False

Question bank

1. Which of the following behavioral factors is most strongly associated with the development of non-communicable diseases?

- A. Regular physical activity**
- B. Balanced diet**
- C. Tobacco smoking**
- D. Adequate hydration**

Correct Answer: C. Tobacco smoking

2. WHO recommends how many minutes of moderate-intensity physical activity per week for adults?

- A. 60 minutes**
- B. 150–300 minutes**
- C. 500 minutes**
- D. 30 minutes only**

Correct Answer: B. 150–300 minutes

3. Which of the following is the causative agent of Typhoid fever?

- A. *Salmonella typhi***
- B. *Shigella dysenteriae***
- C. *Vibrio cholerae***
- D. *Escherichia coli***

Correct Answer: A. *Salmonella typhi*

4. Which disease is transmitted via respiratory droplets and preventable by vaccine?

- A. Hepatitis C**
- B. Measles**

C. Brucellosis

D. Tetanus

Correct Answer: B. Measles

5. Which of the following statements about Hepatitis B is correct?

A. It is transmitted primarily through contaminated food and water.

B. It is a viral infection that affects the respiratory system.

C. It can be prevented by a safe and effective vaccine.

D. It is caused by a bacterial pathogen.

Correct Answer: C. It can be prevented by a safe and effective vaccine.

7. Which of the following statements about rabies is correct?

A. Rabies is transmitted through contaminated food and water.

B. Rabies is a fungal infection affecting the lungs.

C. Rabies is a viral disease transmitted through animal bites.

D. Rabies can be cured after symptoms appear.

Correct Answer: C. Rabies is a viral disease transmitted through animal bites.

8. Brucellosis is primarily transmitted through:

A. Mosquito bites

B. Contaminated water

C. Unpasteurized dairy and animal contact

D. Airborne droplets

Correct Answer: C. Unpasteurized dairy and animal contact

9. Cholera is transmitted through:

A. Airborne droplets

B. Contaminated blood

C. Contaminated water and food

D. Direct skin contact

Correct Answer: C. Contaminated water and food

9. Which of the following is a vector-borne disease?

A. Rabies

B. Malaria

C. Tetanus

D. Hepatitis A

Correct Answer: B. Malaria

10. Which of the following best defines incidence in epidemiology?

A. The total number of existing cases of a disease in a population

B. The number of deaths caused by a disease in a given time

C. The number of new cases of a disease occurring in a population during a specific period

D. The proportion of people exposed to a risk factor

Correct Answer: C. The number of new cases of a disease occurring in a population during a specific period

** True/False Questions**

1. A balanced diet rich in fruits, vegetables, and whole grains helps prevent non-communicable diseases.

True

2. Trans fats are recommended for heart health.

False

3. Physical inactivity is a modifiable risk factor for diabetes and hypertension.

True

4. Adults need only 30 minutes of physical activity per month to maintain health.

False

5. Hepatitis A is a foodborne disease transmitted through contaminated food and water.

True

6. Foodborne diseases are not preventable through hygiene and cooking practices.

False

7. Influenza and pertussis are examples of droplet-borne diseases.

True

8. Droplet-borne diseases can be transmitted through contaminated food.

False

9. Sampling is used to select a representative group from a population for study.

True

10. Prevalence measures the number of new cases of a disease in a population.

False

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- ***Academic records***
- January 2020: M.SC degree in Public Health and Community Medicine from Assiut University, Faculty of medicine, signed “Excellent “.
- September 2014: M.B.B.Ch. degree in medicine & surgery from Assiut University, Faculty of medicine, signed “Excellent with honor”.
- ***Publications***

- Ebrahim OS, Sayed HA, Rabei S, Hegazy N. Perceived stress and anxiety among medical students at Helwan University: A cross-sectional study. J Public Health Res. 2024 Feb 3;13(1):22799036241227891.
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- Ebrahim, O.S., Al-Attar, G.S.T., Gabra, R.H. et al. Stigma and burden of mental illness and their correlates among family caregivers of mentally ill patients. J. Egypt. Public. Health. Assoc. 95, 31 (2020).
<https://doi.org/10.1186/s42506-020-00059-6>

Skills

- *Coordination with Department Chair to ensure curriculum and course material met university guidelines.*
- *Planning and preparing for giving lectures*
- *Checking and assessing student's work*
- *Encouraging personal development through tutorial work*
- *Invigilating examinations, attending staff meetings*
- *Writing research proposals, papers and other publications*
- *Supervising research students*

- *Encouraging team work among students.*

Workshops

- Health Impact assessment (12th July 2025).
- Ethical and regulatory issues in scientific research (21th September, 2021)
- Meta-analysis basic concepts seminar (14th September 2021)
- Access to legal information through the internet (16th January 2021)
- Guidelines to be a competitive applicant to win a Scholarship (20 January 2021)
- Tips for scientific publishing (3rd October 2020)
- On academic integrity, steps to avoid plagiarism (28th September 2020)
- Health system preparedness the upcoming public health challenge (8th -9th October 2018).
- Trainer of the workshop on organizing references using Mendeley software (23th -24th April, 2017).
- Using high impact presentations and policy brief to communicate research results to policy makers. (12th -13th October, 2016).
- Organizing references using Mendeley Software (25th -26th October, 2016).
- Evidence Based Medicine (2nd August, 2016).
- Writing medical reports (4th August, 2016).
- Infection control (6th -7th August, 2016)

Courses

- Medical Korean Academy e-Class Hospital Management (8th June 2025)
- Global disease masterclass; global disease distribution – Imperial College of London (10th June 2023)
- Effective time management- Alfaisal University KLD (30th November 2022)
- Non-communicable diseases in humanitarian settings – University of Copenhagen (30th October 2022)
- Epidemiology the basic science of Public Health - The University of North Carolina at Chapel Hill (9th October 2022)
- Principles of Epidemiology- CDC (September to October 2021)
- Surgical Suturing (10th January, 2016).
- Cardio-Pulmonary Resuscitation (17th-19th January, 2016).
- Ethical, Law & Administrative Considerations in Medical Practice (20th-21th January, 2016).